

106 Geometry Problems

From the AwesomeMath Summer Program

Problems

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Abbreviations and Notation

Notation of geometrical elements

$\angle BAC$	convex angle by vertex A
$\angle(p, q)$	directed angle between lines p and q
$\angle BAC \equiv \angle B'AC'$	angles BAC and $B'AC'$ coincide
AB	line through points A and B , distance between points A and B
$\overline{AB}$	directed segment from point A to point B
$X \in AB$	X lies on the line AB
$X = AC \cap BD$	X is the intersection of the lines AC and BD
$\triangle ABC$	triangle ABC
$[ABC]$	area of $\triangle ABC$
$[A_1 \dots A_n]$	area of polygon $A_1 \dots A_n$
$AB \parallel CD$	lines AB and CD are parallel
$AB \perp CD$	lines AB and CD are perpendicular
$p(X, \omega)$	power of point X with respect to circle ω
$\triangle ABC \cong \triangle DEF$	triangles ABC and DEF are congruent (in this order of vertices)
$\triangle ABC \sim \triangle DEF$	triangles ABC and DEF are similar (in this order of vertices)

Notation of triangle elements

a, b, c	sides or side lengths of $\triangle ABC$
$\angle A, \angle B, \angle C$	angles by vertices A, B , and C of $\triangle ABC$
s	semiperimeter
x, y, z	expressions $\frac{1}{2}(b+c-a)$, $\frac{1}{2}(c+a-b)$, $\frac{1}{2}(a+b-c)$
r	inradius
R	circumradius
K	area
h_a, h_b, h_c	altitudes in $\triangle ABC$
m_a, m_b, m_c	medians in $\triangle ABC$
l_a, l_b, l_c	angle bisectors (segments) in $\triangle ABC$
r_a, r_b, r_c	exradii in $\triangle ABC$

Abbreviations

AMC10	American Mathematics Contest 10
AMC12	American Mathematics Contest 12
AIME	American Invitational Mathematics Examination
USAJMO	United States of America Junior Mathematical Olympiad
USAMO	United States of America Mathematical Olympiad
USA TST	United States of America IMO Team Selection Test
MEMO	Middle European Mathematical Olympiad
IMO	International Mathematical Olympiad

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Chapter 1

Foundations of Geometry

Preliminaries

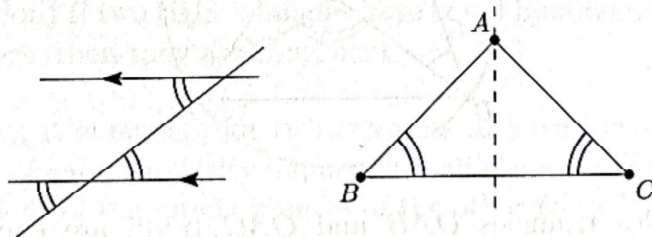
We begin our voyage to the fascinating world of classical geometry by reviewing some elementary facts.

Basic Angles

We state the following:

- Vertical angles are equal.
- A line subtends the same angle with any two parallel lines. In other words, alternate angles are equal.
- In triangle ABC we have $AB = AC$ if and only if $\angle B = \angle C$.

The last two parts of the previous statement need to be taken seriously. The second one offers an efficient way to deal with parallel lines and the third one is one of the very few which translates angles into distances and vice versa.

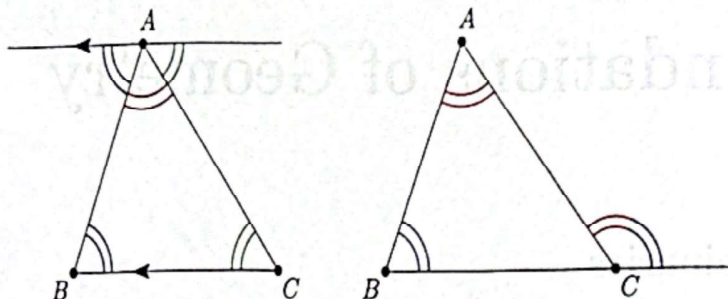


We are ready to prove the universally known theorem on the sum of internal angles in a triangle. In addition, we prove a slight extension, which often offers tiny but pleasant shortcuts in angle calculations.

Proposition 1.1. Let ABC be a triangle with angles $\angle A$, $\angle B$, $\angle C$. Then:

- (a) $\angle A + \angle B + \angle C = 180^\circ$.
 (b) The external angle by vertex C equals $\angle A + \angle B$.

Proof. For (a) draw a line through point A parallel with BC . Since the three angles by vertex A add up to 180° , we arrive at the result by using alternate angles.

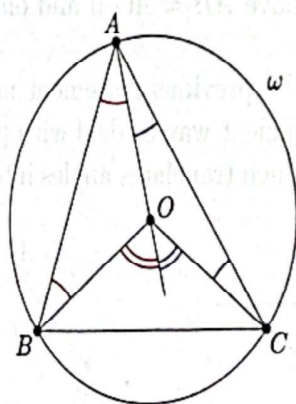


In order to prove (b) we just note that the external angle by vertex C is supplementary to $\angle C$ as well as the sum $\angle A + \angle B$ (by part (a)). $\square$

Also, we know all it takes to prove the Inscribed Angle Theorem, which will later form our understanding of circles.

Theorem 1.2 (Inscribed Angle Theorem). Let BC be a chord of a circle ω centered at O and let $A \in \omega$, $A \neq B, C$. Then the inscribed angle BAC corresponding to arc BC equals one half of the central angle corresponding to the same arc.

Proof. Assume first O lies inside triangle ABC .



From isosceles triangles OAB and OAC (radii are equal!) we infer $\angle OAB = \angle OBA$ and $\angle OAC = \angle OCA$. Then if we extend ray AO beyond O we can find $\angle BOC$ as sum of two external angles. We see that

$$\angle BOC = 2\angle BAO + 2\angle OAC = 2\angle BAC$$

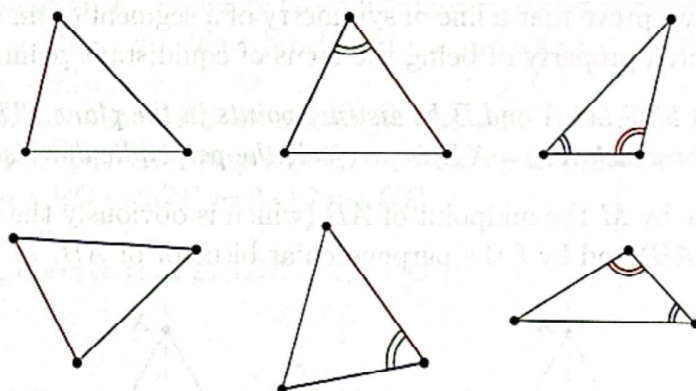
which is exactly what we wanted.

The case when O lies outside or on the boundary of triangle ABC is treated in the same fashion with a few of the additions becoming subtractions. $\square$

Triangle Congruence and Similarity

Informally, we say that two triangles are congruent if they have the same shape and size. Of course, once two triangles are congruent, their corresponding parts (sides, angles, altitudes, ...) are equal. For proving congruence, we have the following criteria:

- (SSS criterion) If three pairs of sides of two triangles are equal in length, then the triangles are congruent.
- (SAS criterion) If two pairs of sides of two triangles are equal in length, and the included angles are equal, then the triangles are congruent.
- (ASA criterion) If two pairs of angles of two triangles are equal, and two corresponding sides are equal in length, then the triangles are congruent.



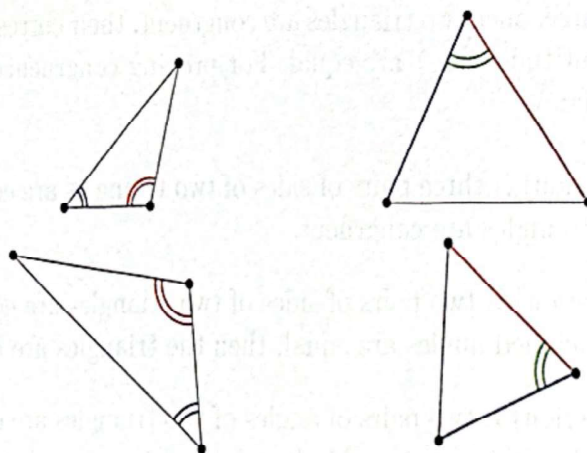
And finally, one criterion which was designed especially for right triangles.

- (HL criterion) If two right triangles have equal hypotenuses and one pair of equal legs, then they are congruent.

For similarity, it is enough for two triangles to have the same shape (i.e. internal angles). Again, similarity implies that all elements of one triangle are just scaled versions of the same elements of the other triangle. Therefore, the ratio of lengths of any corresponding segments is constant. Namely, it is the factor of similarity.

$\square$ The similarity criteria are the following:

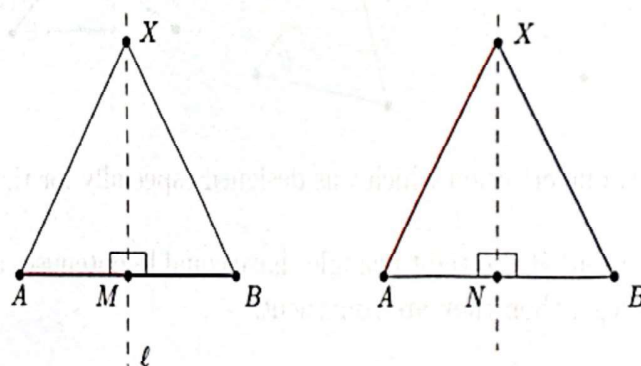
- (AA criterion) If two angles of one triangle are congruent to two angles of another triangle, then the triangles are similar.
- (SAS criterion) If an angle of one triangle is congruent to the corresponding angle of another triangle and the sides that include this angle are proportional, then the two triangles are similar.



Congruence is most frequently used to give rigorous proofs for very natural claims. Here we prove that a line of symmetry of a segment or an angle indeed has the expected property of being the locus of equidistant points.

Proposition 1.3. *Let A and B be distinct points in the plane. Then the locus of points X for which $XA = XB$ is precisely the perpendicular bisector of AB .*

Proof. Denote by M the midpoint of AB (which is obviously the only satisfying point on AB) and by ℓ the perpendicular bisector of AB .

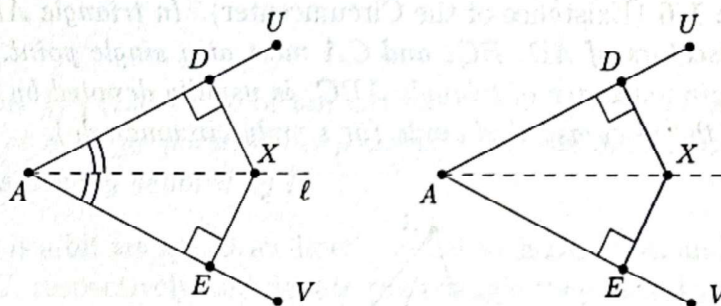


Now if $X \in \ell$ the right triangles AMX and BMX are congruent (SAS: $\angle AMX = \angle BMX = 90^\circ$, $AM = BM$, and XM they have in common) and so $AX = BX$.

On the other hand if $AX = BX$, then let N be the foot of perpendicular from X to AB . Now $\triangle ANX \cong \triangle BNX$ (HL) and thus $AN = NB$ which implies $X \in \ell$. $\square$

Proposition 1.4. Rays AU and AV form an angle. The locus of points X which have the same distance from the rays AU and AV and lie inside angle UAV is precisely the bisector of $\angle UAV$.

Proof. Let D and E be the projections of X onto AU and AV , respectively, and let ℓ be the bisector of angle UAV . If $X \in \ell$, then $\triangle ADX \cong \triangle AEX$ (ASA), hence $XD = XE$.

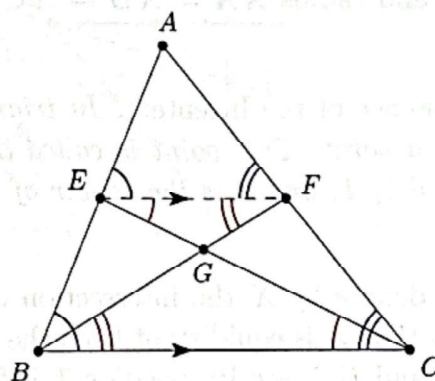


Conversely, if $XD = XE$, we have $\triangle ADX \cong \triangle AEX$ (HL), from which it follows that $\angle XAD = \angle XAE$ and so $X \in \ell$. $\square$

Unlike congruence, similarity has much more striking applications. One of them is that medians divide each other in the ratio $2 : 1$.

Proposition 1.5. Let ABC be a triangle and let E and F be the midpoints of the sides AB and AC , respectively. Denote by G the intersection of BF and CE . Then $BG = 2GF$ and $CG = 2GE$.

Proof. First, observe that $\triangle AEF \sim \triangle ABC$ (SAS).

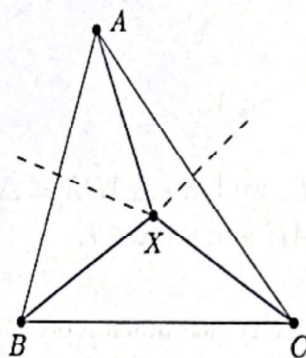


Since the factor of similarity is 2 it follows that $EF = \frac{1}{2}BC$. Moreover, we have $\angle FEA = \angle CBA$, thus $EF \parallel BC$. But then $\angle BCE = \angle CEF$ and we find that $\triangle BCG \sim \triangle FEG$ (AA). Since $EF = \frac{1}{2}BC$, the factor of similarity is $\frac{1}{2}$ and we arrive at the desired equalities $BG = 2GF$ and $CG = 2GE$. $\square$

First Triangle Centers

Despite being such a simple object, the triangle hides perhaps an infinite number of surprising results, many of which are connected to some of its important points. Those are called triangle centers and nowadays over five thousand of them are recognized. Luckily, in olympiad math, it is usually enough to be acquainted with just a small fraction.

Proposition 1.6 (Existence of the Circumcenter). *In triangle ABC the perpendicular bisectors of AB , BC , and CA meet at a single point. This point is called the circumcenter of triangle ABC , is usually denoted by O , and it is the center of the circumscribed circle (or simply circumcircle).*



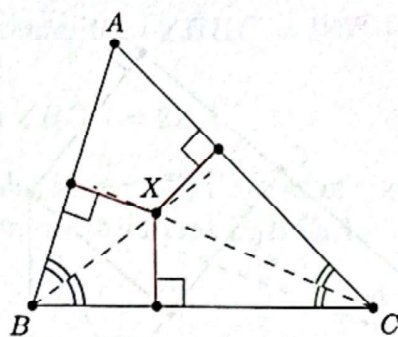
Proof. Let X be the intersection of the perpendicular bisectors of AB and AC . From this we learn $XA = XB$ and $XA = XC$, which gives us $XB = XC$ and this implies that X lies on the perpendicular bisector of BC (if in doubts, see Proposition 1.3).

We have proved that all perpendicular bisectors pass through X . Of course, a circle with center X and radius $XA = XB = XC$ is the circumcircle of triangle ABC . $\square$

Proposition 1.7 (Existence of the Incenter). *In triangle ABC the internal angle bisectors meet at a point. This point is called the incenter of triangle ABC , is usually denoted by I , and it is the center of the incircle of triangle ABC .*

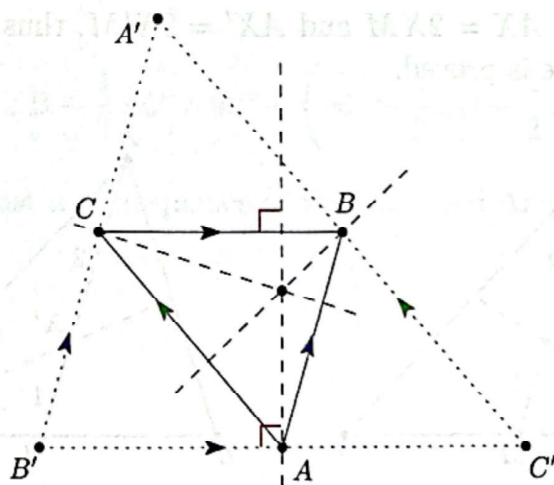
Proof. As expected we denote by X the intersection of the bisectors of $\angle B$ and $\angle C$. Then we know that X is equidistant from the sides AB and BC and also from the sides AC and BC (see Proposition 1.4 if necessary). It follows that X is also equidistant from AB and AC . In other words, it lies on the A -angle bisector. We have found a common point of all three internal angle bisectors.

The circle centered at X having for its radius the common distance from X to the lines BC , CA , and AB is then the incircle of triangle ABC . $\square$



Proposition 1.8 (Existence of the Orthocenter). *In triangle ABC the altitudes meet at a single point. This point is called the orthocenter of triangle ABC and is usually denoted by H .*

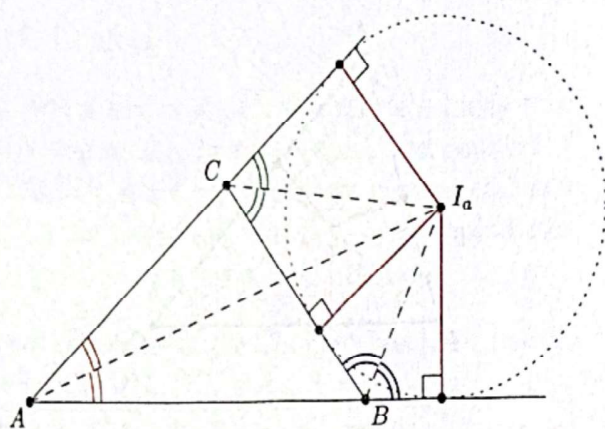
Proof. This is a bit tricky! Draw lines parallel with BC , CA , and AB through A , B , and C , respectively and denote the triangle they form by $A'B'C'$ (with $A'B' \parallel AB$ and likewise for the others). The triangles ABC , $A'CB$, $CB'A$, and BAC' are then all congruent (SAS) and in particular A is the midpoint of $B'C'$ and symmetrically for B and C . This implies that the A -altitude in triangle ABC coincides with the perpendicular bisector of $B'C'$ (both of them being perpendicular to $B'C' \parallel BC$ and passing through A). Since the perpendicular bisectors in triangle $A'B'C'$ are concurrent (see Proposition 1.6), so are the altitudes in triangle ABC .



□

Another triplet of circles bound to a triangle are the excircles. They are in many ways analogous to the incircle and possess numerous remarkable properties.

Proposition 1.9 (Existence of the Excenter). *In triangle ABC the A -angle bisector and the bisectors of external angle B and C meet at a point. This*

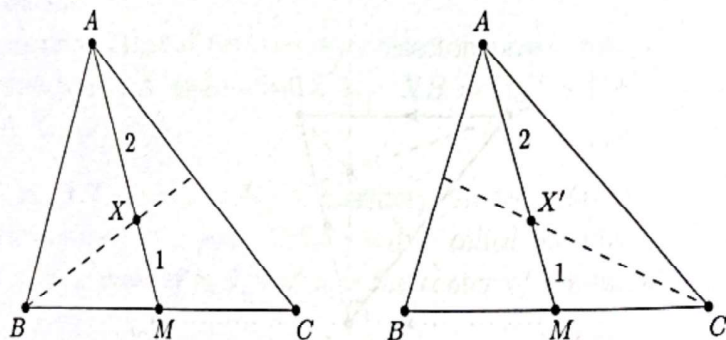


point is called the *A-excenter* of triangle ABC , is usually denoted by I_a and it is the center of the *A-excircle* (circle tangent to the side BC and to the extended sidelines AB and AC). Similarly, we define points I_b and I_c .

Proof. Do it yourself! □

Proposition 1.10 (Existence of the Centroid). *In triangle ABC the medians meet at a point. This point is called the centroid of triangle ABC and is usually denoted by G .*

Proof. Let X be the intersection of the median AM with the B -median and let X' be the intersection of AM with the C -median. From Proposition 1.5 we know that both $AX = 2XM$ and $AX' = 2X'M$, thus inevitably $X = X'$ and the concurrence is proved.



□

One may be tempted to believe the following result is not worth remembering as it is so easy to derive. But this would be a terrible mistake! In fact, many connections are revealed if one knows this without thinking!

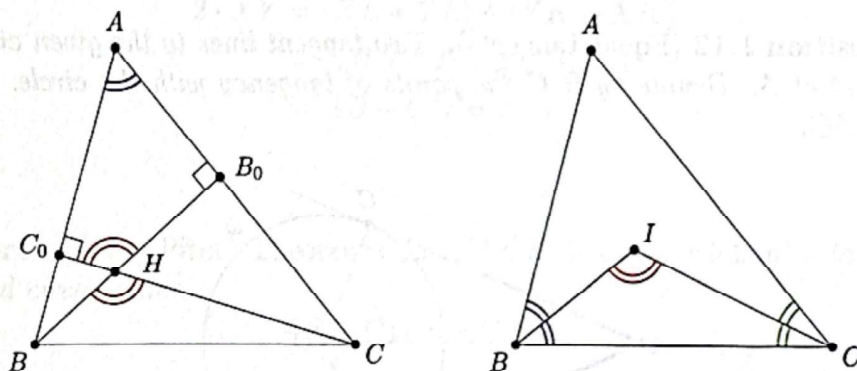
Proposition 1.11. *Let ABC be a triangle with orthocenter H , incenter I , and circumcenter O . Then:*

(a) If triangle ABC is acute, then $\angle BHC = 180^\circ - \angle A$.

(b) $\angle BIC = 90^\circ + \frac{1}{2}\angle A$.

(c) If $\angle A$ is acute, then $\angle BOC = 2\angle A$.

Proof. In (a) we denote by B_0 and C_0 the feet of the altitudes from B and C , respectively, and focus on quadrilateral B_0HC_0A .



Since the sum of internal angles in a quadrilateral is 360° and $\angle HB_0A = \angle HC_0A = 90^\circ$, the remaining two angles add up to 180° . In other words:

$$\angle BHC = \angle C_0HB_0 = 180^\circ - \angle A.$$

For (b) we angle-chase in triangle BIC . Since BI and CI are angle bisectors, we have

$$\angle BIC = 180^\circ - \frac{1}{2}\angle B - \frac{1}{2}\angle C = 90^\circ + \left(90^\circ - \frac{1}{2}\angle B - \frac{1}{2}\angle C\right) = 90^\circ + \frac{1}{2}\angle A.$$

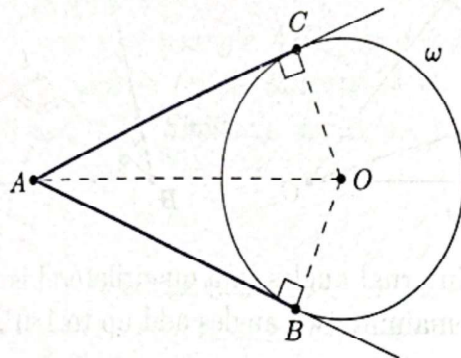
Finally, (c) is just a consequence of the Inscribed Angle Theorem. $\square$

Metric Relations

Equal Tangents

We start with a very simple method, which on the other hand has many nontrivial applications and appears over and over again in contests. We will just play with equal segments.

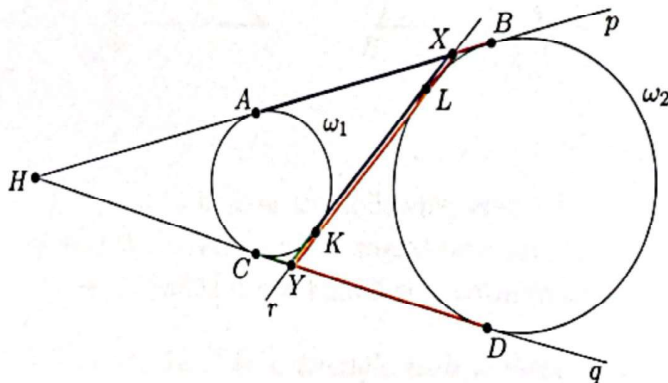
Proposition 1.12 (Equal Tangents). *Two tangent lines to the given circle ω intersect at A . Denote by B, C the points of tangency with the circle. Then $AB = AC$.*



Proof. Let O be the center of ω . Then $\angle OBA = 90^\circ = \angle OCA$ and $OB = OC$. Moreover, the right triangles OAB and OAC share the hypotenuse OA , thus they are congruent (HL). The result follows. $\square$

Proposition 1.13. *Let p, q be common external tangents of circles ω_1, ω_2 . Denote by A, B the points of tangency of p with ω_1 and ω_2 , respectively, and by C, D the points of tangency of q with ω_1 and ω_2 , respectively. Then:*

- (a) $AB = CD$.
- (b) *If the two circles are nonintersecting and their common internal tangent r intersects p and q at points X, Y , respectively, we have $AB = CD = XY$.*



Proof. For part (a), if $AB \parallel CD$ then $ABCD$ is a rectangle and we are done. Otherwise, let $AB \cap CD = H$. Now by Equal Tangents we have $HA = HC$ and $HB = HD$. Subtracting gives the result.

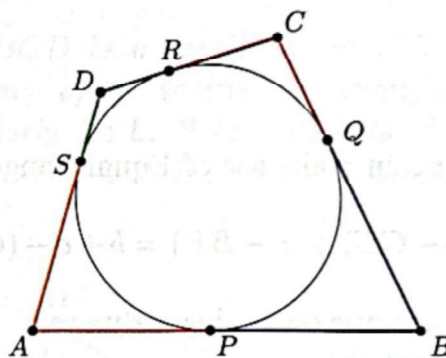
In part (b), denote by K, L the points of tangency of r_1 and ω_1, ω_2 , respectively. Now using part (a) and Equal Tangents several times, we obtain

$$\begin{aligned} 2 \cdot XY &= (XL + YL) + (YK + XK) \\ &= XB + YD + YC + XA \\ &= AB + CD = 2 \cdot AB. \end{aligned}$$

□

Theorem 1.14 (Pitot¹ Theorem). *Let $ABCD$ be a quadrilateral with an inscribed circle. Then*

$$AB + CD = BC + DA.$$



Proof. Denote by P, Q, R, S the points of tangency of the inscribed circle with the sides AB, BC, CD, DA , respectively. Equal Tangents give

$$AB + CD = AP + BP + CR + DR = AS + BQ + CQ + DS = BC + DA$$

and we are done. □

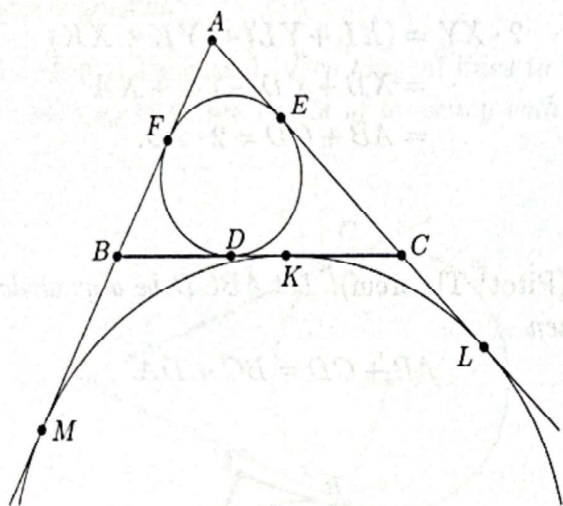
In fact, the condition $AB + CD = BC + DA$ is also sufficient for a quadrilateral to have an inscribed circle. Can you find the proof?

We are ready to prove an important fact from the geometry of a triangle.

Proposition 1.15. *Let ABC be a triangle with semiperimeter s . Denote by D, E, F the points of tangency of the incircle with the sides BC, CA, AB , respectively. Also let the A -excircle touch the lines BC, CA, AB at points K, L, M , respectively. Then the following holds:*

¹Henri Pitot (1695–1771) was a French hydraulic engineer.

- (a) $2 \cdot AE = 2 \cdot AF = -a + b + c$, $2 \cdot BD = 2 \cdot BF = a - b + c$,
 $2 \cdot CD = 2 \cdot CE = a + b - c$.
 (b) $2AL = 2AM = a + b + c$, in other words $AL = AM = s$.
 (c) Points K and D are symmetric with respect to the midpoint of BC .



Proof. For part (a), we again make use of Equal Tangents. Namely

$$2 \cdot AE = AE + AF = (b - CE) + (c - BF) = b + c - (CD + BD) = b + c - a.$$

The other two relations are proved analogously.

Similarly, we obtain part (b):

$$2 \cdot AL = AL + AM = (b + CL) + (c + BM) = b + c + (CK + BK) = a + b + c.$$

To prove part (c), it suffices to show that $BD = CK$ and indeed using parts (a) and (b), we easily derive

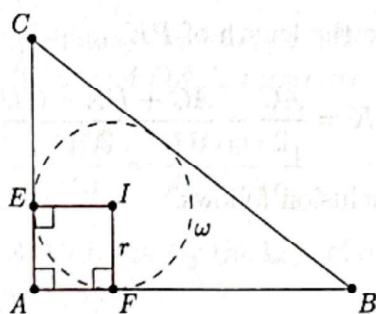
$$2 \cdot BD = a - b + c = 2(s - b) = 2(AL - AC) = 2 \cdot CL = 2 \cdot CK$$

and thus conclude the proof. $\square$

This result immediately gives a convenient formula for the inradius of a right triangle.

Proposition 1.16. In right triangle ABC , where $\angle A = 90^\circ$, denote by r the radius of its incircle ω . Then

$$r = \frac{AB + AC - CB}{2}.$$

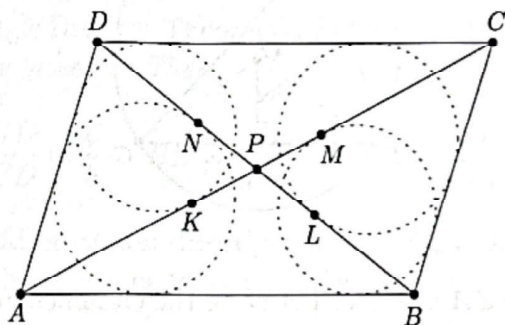


Proof. Denote by E and F the points of contact of ω with AC and AB , respectively, and by I the center of ω .

We take a look at quadrilateral $AFIE$. It has three right angles and equal pairs of adjacent sides $AF = AE$ (Equal Tangents) and $IE = IF$, which makes it a square. Thus $r = AE$ and the previous proposition gives the desired

$$r = AE = \frac{AB + AC - CB}{2}. \quad \square$$

Example 1.1. Let $ABCD$ be a parallelogram with $AB > BC$. Let K, M be the points of tangency of the incircles of triangles ACD and ABC with AC , respectively. Similarly, let L, N be the points of tangency of the incircles of triangles BCD and ABD with BD , respectively. Prove that $KLMN$ is a rectangle.



Proof. Let P be the intersection of diagonals in $ABCD$. First, we observe that triangles ABD and CDB are congruent (ASA) and symmetric with respect to P . Therefore, we have $PN = PL$. Similarly, we deduce that $PK = PM$, thus the diagonals in $KLMN$ bisect each other and this quadrilateral is a parallelogram.

To prove that it is in fact a rectangle, it suffices to show $NL = KM$ or equivalently $PN = PK$. By Proposition 1.15(a) applied to triangle ABD we get

$$PN = \frac{DB}{2} - DN = \frac{DB}{2} - \frac{DB + DA - AB}{2} = \frac{AB - DA}{2}.$$

Analogously, we calculate the length of PK :

$$PK = \frac{AC}{2} - AK = \frac{AC}{2} - \frac{AC + DA - CD}{2} = \frac{CD - DA}{2}.$$

Since $AB = CD$, the conclusion follows. $\square$

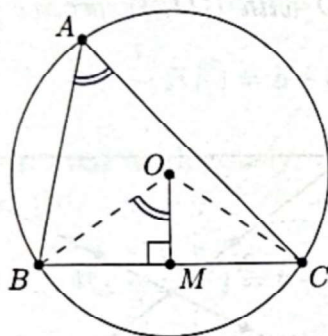
The Law of Sines

Now we will discuss one of the most fundamental theorems from the triangle geometry, the Law of Sines. In fact, the method of trigonometric computation is a very powerful technique and for an aspiring contestant it is a must to know. One of the reasons why the Law of Sines is so useful is the well-known fact that $\sin x = \sin(180^\circ - x)$.

Theorem 1.17 (The Extended Law of Sines). *Let ABC be a triangle. Then*

$$\frac{a}{\sin \angle A} = \frac{b}{\sin \angle B} = \frac{c}{\sin \angle C} = 2R,$$

where R is the circumradius of triangle ABC .



Proof. Assume that $\angle A$ is acute. Let O be the circumcenter of triangle ABC and M the midpoint of BC . Then $\angle BOC = 2\angle A$ as it is a central angle. Triangle OBC is isosceles, so OM bisects $\angle BOC$ and $\angle BOM = \angle A$. From right triangle BOM we obtain

$$\sin \angle A = \frac{\frac{1}{2}a}{R},$$

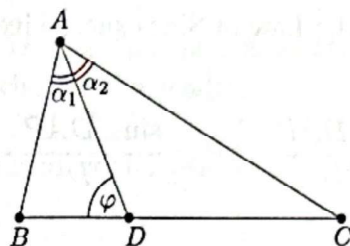
implying the result. The case when $\angle A$ is not acute can be proved analogously using the fact that $\sin \angle A = \sin(180^\circ - \angle A)$. Details are left for the reader. $\square$

The following lemma provides a useful shortcut whenever we are dealing with ratios in adjacent triangles.

Proposition 1.18 (Ratio Lemma). *In triangle ABC , let $D \in BC$ and denote by α_1 and α_2 the angles BAD and DAC , respectively. Then*

$$\frac{BD}{DC} = \frac{AB \cdot \sin \alpha_1}{AC \cdot \sin \alpha_2}.$$

Proof. Denote the angle ADB by φ . By the Law of Sines in adjacent triangles ADB and ADC , we obtain



$$\frac{BD}{\sin \alpha_1} = \frac{AB}{\sin \varphi}, \quad \frac{CD}{\sin \alpha_2} = \frac{AC}{\sin(180^\circ - \varphi)} = \frac{AC}{\sin \varphi}.$$

Now dividing the two relations yields the result. $\square$

As an immediate corollary we obtain the well-known Angle Bisector Theorem.

Theorem 1.19 (Angle Bisector Theorem). *In triangle ABC let AD , $D \in BC$, be the internal angle bisector. Then*

$$\frac{BD}{CD} = \frac{c}{b}, \quad BD = \frac{ac}{b+c}, \quad CD = \frac{ab}{b+c}.$$

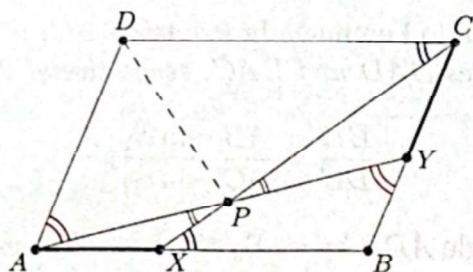
Proof. The first relation comes directly from the Ratio Lemma as $\alpha_1 = \alpha_2$. The other two are the solutions to system of equations

$$\frac{BD}{CD} = \frac{c}{b} \quad \text{and} \quad BD + CD = a. \quad \square$$

The next example illustrates typical use of the Law of Sines.

Example 1.2 (Germany 2003). *Let $ABCD$ be a parallelogram. Let X and Y be points on the sides AB and BC , respectively, such that $AX = CY$. Prove that the intersection of lines AY and CX lies on the angle bisector of $\angle ADC$.*

Proof. Denote the intersection of AY and CX by P . First, we make use of parallel lines to see that $\angle DAP = 180^\circ - \angle PYC$ and $\angle DCP = 180^\circ - \angle PXA$.



Now we are ready to use the Law of Sines successively in triangles APD and APX to obtain

$$\sin \angle ADP = \frac{\sin \angle DAP}{PD} \cdot AP = \frac{\sin \angle DAP}{PD} \cdot AX \cdot \frac{\sin \angle PXA}{\sin \angle APX},$$

and similarly

$$\sin \angle CDP = \frac{\sin \angle DCP}{PD} \cdot CP = \frac{\sin \angle DCP}{PD} \cdot CY \cdot \frac{\sin \angle PYC}{\sin \angle CPY}.$$

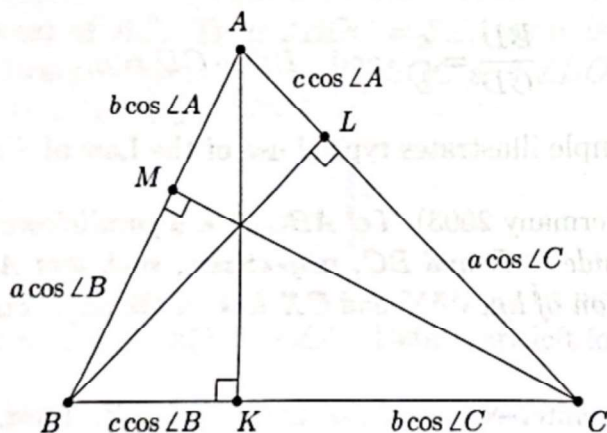
But $AX = CY$, $\angle APX = \angle CPY$, and we already know that $\sin \angle DAP = \sin \angle PYC$ and $\sin \angle DCP = \sin \angle PXA$, hence $\sin \angle ADP = \sin \angle CDP$. Moreover, $\angle ADP + \angle CDP \neq 180^\circ$, thus $\angle ADP = \angle CDP$. $\square$

The Law of Cosines

We present another theorem from basic triangle geometry, the Law of Cosines. Again this theorem is more useful than it might seem at the first glance.

Theorem 1.20 (Law of Cosines). *Let ABC be a triangle. Then*

$$a^2 = b^2 + c^2 - 2bc \cos \angle A.$$



Proof. Assume triangle ABC is acute. Let AK , BL , CM be the altitudes in triangle ABC . Using right triangles yields

$$\begin{aligned} a^2 &= a(BK + KC) = a(c \cos \angle B) + a(b \cos \angle C) \\ &= c(a \cos \angle B) + b(a \cos \angle C) \\ &= c \cdot BM + b \cdot CL = c(c - b \cos \angle A) + b(b - c \cos \angle A) \\ &= c^2 + b^2 - 2bc \cos \angle A. \end{aligned}$$

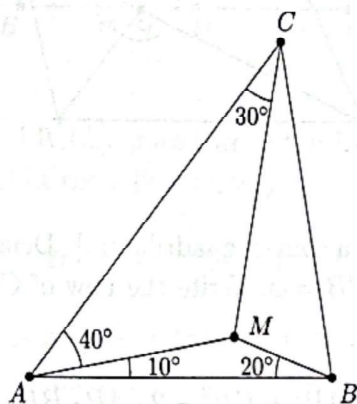
The cases when triangle ABC is right or obtuse can be done analogously and we leave them as an exercise for the reader. $\square$

Corollary 1.21 (Generalized Pythagorean² Theorem). *Let ABC be a triangle. Then:*

- (a) $\angle C < 90^\circ$ if and only if $a^2 + b^2 > c^2$.
- (b) $\angle C = 90^\circ$ if and only if $a^2 + b^2 = c^2$.
- (c) $\angle C > 90^\circ$ if and only if $a^2 + b^2 < c^2$.

Proof. This is a direct consequence of the Law of Cosines and the fact that the function $\cos$ is positive on $(0^\circ, 90^\circ)$, negative on $(90^\circ, 180^\circ)$, and zero for 90° . $\square$

Example 1.3 (USAMO 1996, Titu Andreescu). *Let ABC be a triangle and M an interior point such that $\angle MAB = 10^\circ$, $\angle MBA = 20^\circ$, $\angle MAC = 40^\circ$, and $\angle MCA = 30^\circ$. Prove that triangle ABC is isosceles.*



Proof. Assume without loss of generality $AB = 1$. First note $\angle AMB = 150^\circ$ and $\angle AMC = 110^\circ$. The key is to realize that we can calculate all three side

²Pythagoras of Samos (c. 570–495 BC) was a greek philosopher and mathematician.

lengths. Indeed, using the Law of Sines in triangles AMB and AMC , and the formula $\sin 2x = 2 \sin x \cos x$ yields

$$AM = AB \cdot \frac{\sin 20^\circ}{\sin 150^\circ} = 2 \sin 20^\circ$$

and

$$AC = AM \cdot \frac{\sin 110^\circ}{\sin 30^\circ} = (2 \sin 20^\circ) \cdot 2 \cdot \cos 20^\circ = 2 \sin 40^\circ.$$

From the Law of Cosines in triangle ABC we obtain

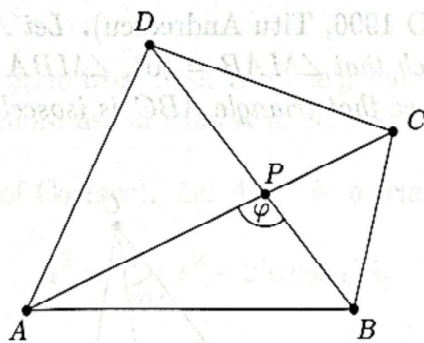
$$\begin{aligned} BC^2 &= 1^2 + (2 \sin 40^\circ)^2 - 2 \cdot 1 \cdot 2 \sin 40^\circ \cos 50^\circ \\ &= 1 + 4 \sin^2 40^\circ - 4 \sin^2 40^\circ = 1. \end{aligned}$$

Thus $AB = BC$ and triangle ABC is isosceles as desired. $\square$

Now we shall present a metric criterion for orthogonality, that is a straightforward application of the Law of Cosines.

Proposition 1.22. *Let AC and BD be two lines in plane. Then, $AC \perp BD$ if and only if*

$$AB^2 + CD^2 = AD^2 + BC^2.$$



Proof. Assume $ABCD$ is a convex quadrilateral. Denote by P the intersection of diagonals and let $\angle APB = \varphi$. Write the Law of Cosines in triangles ABP and CDP :

$$AB^2 = AP^2 + BP^2 - 2 \cdot AP \cdot BP \cdot \cos \varphi,$$

$$CD^2 = CP^2 + DP^2 - 2 \cdot CP \cdot DP \cdot \cos \varphi.$$

Adding the equations we obtain

$$AB^2 + CD^2 = AP^2 + BP^2 + CP^2 + DP^2 - 2(CP \cdot DP + AP \cdot BP) \cos \varphi.$$

Similarly, we add the Law of Cosines from triangles BCP and DAP , keeping in mind that $\cos x = -\cos(180^\circ - x)$. We get

$$BC^2 + DA^2 = BP^2 + CP^2 + DP^2 + AP^2 + 2(BP \cdot CP + DP \cdot AP) \cos \varphi.$$

Comparing, we see that $AB^2 + CD^2 = BC^2 + DA^2$ holds if and only if

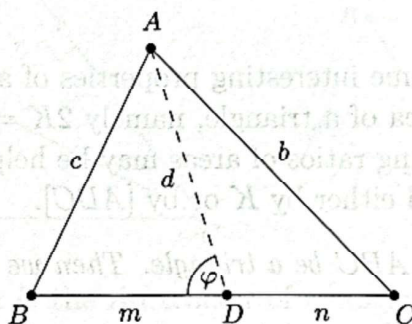
$$(CP \cdot DP + AP \cdot BP + BP \cdot CP + DP \cdot AP) \cos \varphi = 0.$$

Since the former quantity is positive, this may only happen if $\cos \varphi = 0$ i.e. if $\varphi = 90^\circ$.

The case when $ABCD$ is not convex is handled analogously and the cases when $ABCD$ is not a quadrilateral are even easier. $\square$

Theorem 1.23 (Stewart's³ theorem). *In triangle ABC let D lie on the side BC . Denote the distances BD, DC , and AD by m, n , and d , respectively. Then*

$$a(d^2 + mn) = b^2m + c^2n.$$



Proof. Denote the angle ADB by φ and write the Law of Cosines in adjacent triangles ABD and ADC in the following way:

$$c^2 = d^2 + m^2 - 2md \cos \varphi, \quad b^2 = d^2 + n^2 + 2nd \cos \varphi.$$

Multiplying the first relation by n and the second by m so that we can easily eliminate the cosine just by adding the equations, we obtain

$$c^2n + b^2m = d^2n + m^2n + d^2m + n^2m = (m+n)(d^2 + mn) = a(d^2 + mn),$$

and we are done. $\square$

³Matthew Stewart (1719–1785) was a Scottish mathematician and minister of religion.

Corollary 1.24. In triangle ABC let M be the midpoint of BC and let AD , $D \in BC$, be the internal angle bisector. Then

$$m_a^2 = AM^2 = \frac{b^2 + c^2}{2} - \frac{a^2}{4}, \quad l_a^2 = AD^2 = bc \left(1 - \left(\frac{a}{b+c} \right)^2 \right).$$

Proof. The first relation is immediate as we have $m = n = \frac{1}{2}a$ in Stewart's theorem.

For the second part we recall the Angle Bisector Theorem (see Theorem 1.19) and apply Stewart's theorem to obtain

$$a \left(l_a^2 + \frac{a^2 bc}{(b+c)^2} \right) = \frac{b^2 ac}{b+c} + \frac{c^2 ab}{b+c}.$$

After dividing by a and simplifying the right-hand side we get

$$l_a^2 + \frac{a^2 bc}{(b+c)^2} = bc,$$

which is easily seen to be equivalent to what we are proving. $\square$

Areas

Now we shall develop some interesting properties of areas, starting only with the basic formula for area of a triangle, namely $2K = ah_a = bh_b = ch_c$. Also we will see how calculating ratios of areas may be helpful. Recall that area of triangle ABC is denoted either by K or by $[ABC]$.

Proposition 1.25. Let ABC be a triangle. Then we can calculate its area in the following ways:

- (a) $K = \frac{1}{2}ab \sin \angle C = \frac{1}{2}bc \sin \angle A = \frac{1}{2}ca \sin \angle B = abc/(4R)$.
- (b) $K = \sqrt{s(s-a)(s-b)(s-c)}$ (Heron's⁴ formula).
- (c) $K = rs = r_a(s-a) = r_b(s-b) = r_c(s-c)$.

Proof. For the first part of (a), it suffices to prove (by symmetry) $2K = ab \sin \angle C$, but as $h_a = b \sin \angle C$ this is obvious. Also, by the Extended Law of Sines

$$\frac{1}{2}ab \sin \angle C = \frac{1}{2}ab \frac{c}{2R} = \frac{abc}{4R}.$$

By part (a) we have $16K^2 = 4a^2b^2 \sin^2 \angle C = 4a^2b^2(1 - \cos^2 \angle C)$ and by the Law of Cosines

$$\cos \angle C = \frac{a^2 + b^2 - c^2}{2ab}.$$

⁴Heron of Alexandria (c. 10–70) was an ancient Greek mathematician and engineer.

Putting this together we obtain

$$16K^2 = 4a^2b^2 \left(1 - \frac{(a^2 + b^2 - c^2)^2}{4a^2b^2} \right) = 4a^2b^2 - (a^2 + b^2 - c^2)^2,$$

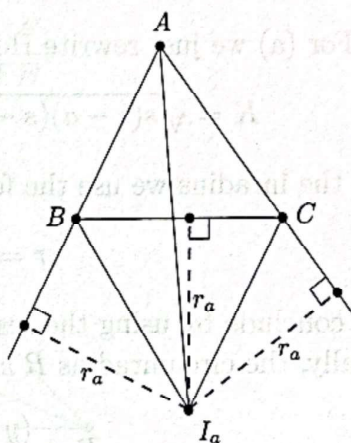
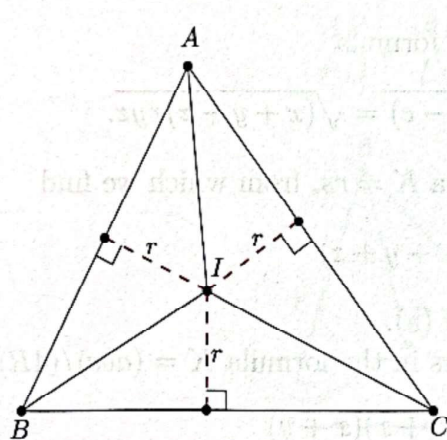
which factors as

$$\begin{aligned} (2ab - a^2 - b^2 + c^2)(2ab + a^2 + b^2 - c^2) &= (c^2 - (a - b)^2)((a + b)^2 - c^2) \\ &= (-a + b + c)(a - b + c)(a + b - c)(a + b + c), \end{aligned}$$

but this is just an equivalent form of Heron's formula so we have proved part (b).

Finally for (c), denote by I the incenter of triangle ABC and observe that

$$K = [BIC] + [CIA] + [AIB] = \frac{1}{2}(ra + rb + rc) = rs.$$



Similarly, denote by I_a the A -excenter of triangle ABC and again observe

$$K = [AI_aC] + [BI_aA] - [BI_aC] = \frac{1}{2}(br_a + cr_a - ar_a) = r_a(s - a).$$

The remaining relations are proved in a similar fashion, so the proof is finished. $\square$

The main application of these area formulas is that they allow us to express some common triangle elements in terms of triangle sides. This is especially convenient if we use the standard notation xyz :

$$x = s - a = \frac{1}{2}(b + c - a), \quad y = s - b = \frac{1}{2}(c + a - b), \quad z = s - c = \frac{1}{2}(a + b - c).$$

These expressions come in fact from Proposition 1.15. Another way to look at x , y , and z is that they are the unique numbers such that

$$a = y + z, \quad b = z + x, \quad c = x + y.$$

Since calculating triangle elements is a common theme in many problems, we will appreciate that the xyz notation simplifies the computations.

Proposition 1.26 (xyz formulas). *In triangle ABC we can find the area K , inradius r , and circumradius R in terms of xyz as follows:*

$$(a) \quad K = \sqrt{(x+y+z)xyz},$$

$$(b) \quad r = \sqrt{\frac{xyz}{x+y+z}},$$

$$(c) \quad R = \frac{(y+z)(z+x)(x+y)}{4\sqrt{xyz(x+y+z)}}.$$

Proof. For (a) we just rewrite Heron's formula

$$K = \sqrt{s(s-a)(s-b)(s-c)} = \sqrt{(x+y+z)xyz}.$$

To find the inradius we use the formula $K = rs$, from which we find

$$r = K/(x+y+z)$$

and we conclude by using the result of (a).

Finally, the circumradius R appears in the formula $K = (abc)/(4R)$, thus

$$R = \frac{(y+z)(z+x)(x+y)}{4K}$$

and again part (a) ensures the rest. $\square$

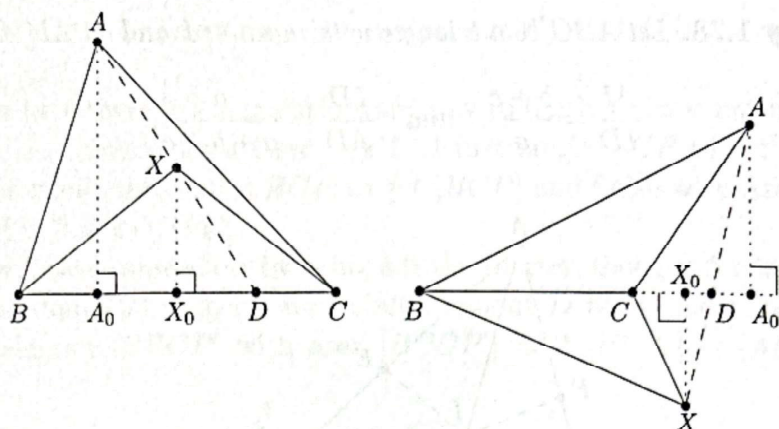
The following lemma is surprisingly handy, because it efficiently transfers ratios of distances to ratios of areas. We will refer to this result as the *Area Lemma*.

Proposition 1.27 (Area Lemma). *Let ABC be a triangle, $D \in BC$ and $X \in AD$. Then*

$$\frac{[BCX]}{[BCA]} = \frac{DX}{DA}.$$

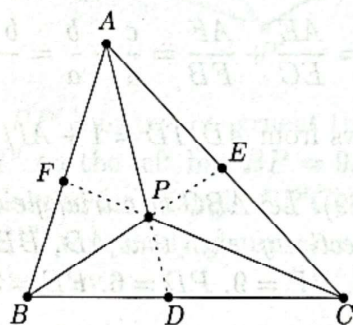
Proof. Drop perpendiculars from A and X to BC and denote their feet by A_0 and X_0 , respectively. The triangles DX_0X and DA_0A are similar (AA). Hence

$$\frac{[BCX]}{[BCA]} = \frac{\frac{1}{2} \cdot BC \cdot XX_0}{\frac{1}{2} \cdot BC \cdot AA_0} = \frac{XX_0}{AA_0} = \frac{DX}{DA}. \quad \square$$



Example 1.4 (van Aubel's⁵ Theorem). In triangle ABC let D, E, F be points on the sides BC, CA, AB , respectively, such that AD, BE , and CF are concurrent at P . Then

$$\frac{AP}{PD} = \frac{AE}{EC} + \frac{AF}{FB}.$$



Proof. By the Area Lemma (Proposition 1.27) the right-hand side equals

$$\frac{[APB]}{[BPC]} + \frac{[APC]}{[BPC]} = \frac{[ABPC]}{[BPC]} = \frac{[ABC]}{[BCP]} - 1.$$

It remains to apply the Area Lemma again, this time on the left-hand side:

$$\frac{AP}{PD} = \frac{AD}{PD} - 1 = \frac{[ABC]}{[BCP]} - 1.$$

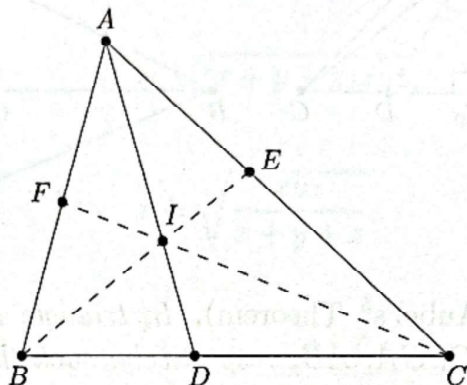
This concludes the proof. $\square$

Note that if we choose E and F to be the midpoints of the sides AC and AB , we obtain another proof that medians divide each other in the ratio $2 : 1$ (see Proposition 1.5). Another notable corollary follows if we take P to be the incenter of triangle ABC .

⁵Henri Hubert van Aubel (1830–1906) was a Dutch professor of mathematics.

Corollary 1.28. Let ABC be a triangle with incenter I and let $AI \cap BC = D$. Then:

$$\frac{AI}{ID} = \frac{b+c}{a} \quad \text{and} \quad \frac{ID}{AD} = \frac{a}{a+b+c}.$$

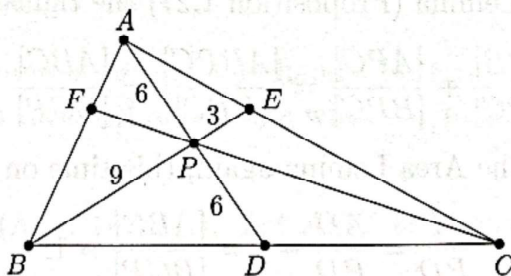


Proof. Let $E = BI \cap AC$ and $F = CI \cap AB$. Then van Aubel's Theorem combined with the Angle Bisector Theorem imply

$$\frac{AI}{ID} = \frac{AE}{EC} + \frac{AF}{FB} = \frac{c}{a} + \frac{b}{a} = \frac{b+c}{a}.$$

The second relation follows from $AD/ID = 1 + AI/ID$. $\square$

Example 1.5 (AIME 1989). Let ABC be a triangle and let D, E, F lie on the sides BC, CA, AB , respectively, such that AD, BE and CF are concurrent at P . Given that $AP = 6$, $BP = 9$, $PD = 6$, $PE = 3$, and $CF = 20$, find the area of triangle ABC .



Proof. By the Area Lemma (see Proposition 1.27) we have

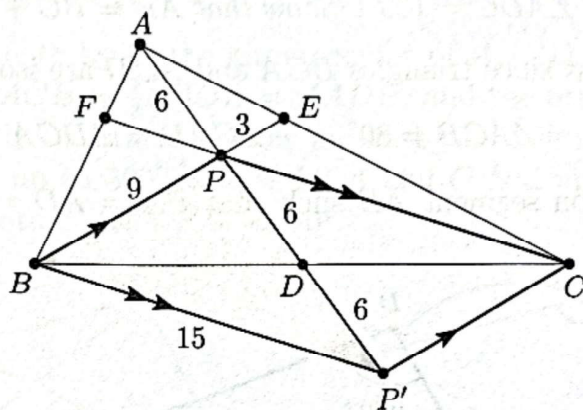
$$\frac{[BPC]}{[ABC]} = \frac{DP}{DA} = \frac{1}{2}, \quad \frac{[CPA]}{[ABC]} = \frac{EP}{EB} = \frac{1}{4}.$$

Since $[ABC] = [BPC] + [CPA] + [APB]$, this implies $[APB] = \frac{1}{4}[ABC]$, and using the Area Lemma again we obtain $FP = \frac{1}{4}CF = 5$ and $CP = 15$. Also,

$[ABP] = [CPA]$ implies (Area Lemma yet again!) that D is the midpoint of BC .

From here there is a straightforward way to finish without much thinking: Use median formula from Corollary 1.24 in triangle BCP to get BC , then Heron's formula in triangle BCP to get $[BCP]$ and finish with Area Lemma to get $[ABC] = 2 \cdot [BPC]$.

We will save some effort by being a little smarter, though. A triangle BPC with a median PD suggests we reflect P about D to P' and thus complete a parallelogram $BPCP'$ with area $[BPCP'] = 2 \cdot [BPC] = [ABC]$. The



parallelogram is split by PP' into two congruent triangles whose side lengths we know – triangle BPP' on the left has $BP = 9$, $PP' = 2 \cdot PD = 12$ and $P'B = CP = 15$ (and likewise for triangle $PP'C$). Instead of using Heron's formula to compute $[BPP']$, we quickly glance at numbers 9, 12, 15, recognize a right triangle and conclude by

$$[ABC] = 2 \cdot [BPP'] = 2 \cdot \frac{1}{2} \cdot 9 \cdot 12 = 108. \quad \square$$

Circles, Angles

As we will see in this section, angles are intimately related to many common geometric configurations. Being able to manipulate them smoothly is therefore a skill one has to learn.

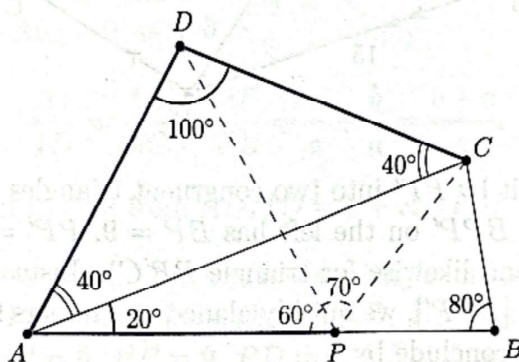
In the following proof we will reach the result by computing various angles in the picture. This technique is called *angle-chasing* and it is probably the most frequently used method in Euclidean geometry.

Example 1.6. Let $ABCD$ be a quadrilateral such that $AB = AC$, $AD = CD$, $\angle BAC = 20^\circ$, and $\angle ADC = 100^\circ$. Show that $AB = BC + CD$.

Proof. Observe that since triangles BCA and ACD are isosceles, we have

$$\angle CBA = \angle ACB = 80^\circ, \quad \angle CAD = \angle DCA = 40^\circ.$$

Let P be a point on segment AB such that $PA = AD = CD$. We aim to prove $BP = BC$.



Note that $\angle PAD = 20^\circ + 40^\circ = 60^\circ$. Triangle PAD has two equal sides AP and AD that subtend angle 60° , so it is equilateral. From this we deduce $PD = AD = CD$, so triangle PDC is isosceles and, since $\angle PDC = 100^\circ - 60^\circ = 40^\circ$, we get $\angle DCP = \angle CPD = 70^\circ$.

Finally, turning our attention to triangle BCP we observe that

$$\angle BPC = 180^\circ - 60^\circ - 70^\circ = 50^\circ$$

which together with $\angle CBP = 80^\circ$ implies $\angle PCB = 50^\circ$, so $PB = BC$ indeed. $\square$

Circles

Angle-chasing would not be as important and powerful if we were not able to employ circles. Fortunately, this is possible thanks to the Inscribed Angle Theorem (Theorem 1.2). Recall that a quadrilateral is called *cyclic* (or *inscribed*) if it can be inscribed in a circle.